

C++ Programming Language

* Why do we need loops

```
void main() {  
    Print('Hello');  
    Print('Hello');  
    Print('Hello');  
    Print('Hello');  
    Print('Hello');  
}
```

```
void main() {  
    for (int i = 1; i <= 5; i++) {  
        Print('Hello');  
    }  
}
```

> for Loop

Syntax:

for (initialization ; condition ; increment / decrement) {

Code

}

a) Step up (1-5)

```
void main() {  
    for (int i = 1; i <= 5; i++) {  
        Print(i);  
    }  
}
```

1
2
3
4
5

b) Step down (5-1)

```
void main() {  
    for (int i = 5; i >= 1; i--) {  
        Print(i);  
    }  
}
```

5
4
3
2
1

c) Even Number (1-10)

```
void main() {  
    for (int i=2; i<=10; i+=2) {  
        print(i);  
    }  
}
```

2
4
6
8
10

d) Odd Number (1-10)

```
void main() {  
    for (int i=1; i<=10; i+=2) {  
        print(i);  
    }  
}
```

1
3
5
7
9

e) Multiple Variables

```
void main() {  
    for (int i=1, j=10; i<j; i++, j--) {  
        print("$i => $j");  
    }  
}
```

1 => 10
2 => 9
3 => 8
4 => 7
5 => 6

2. While Loop

Syntax:

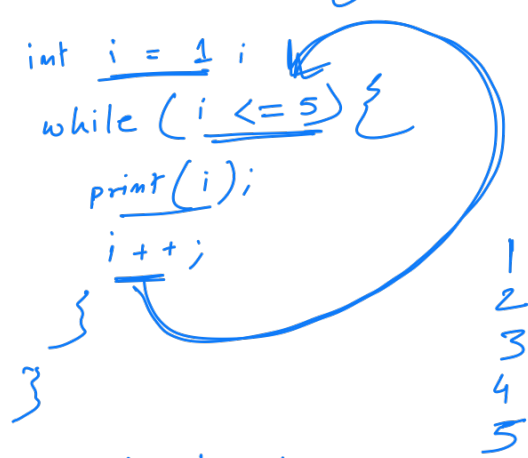
```
while (condition) {
```

code

```
}
```

a) Step UP (1-5)

```
void main() {  
    int i = 1 i  
    while (i <= 5) {  
        print(i);  
        i++;  
    }  
}
```



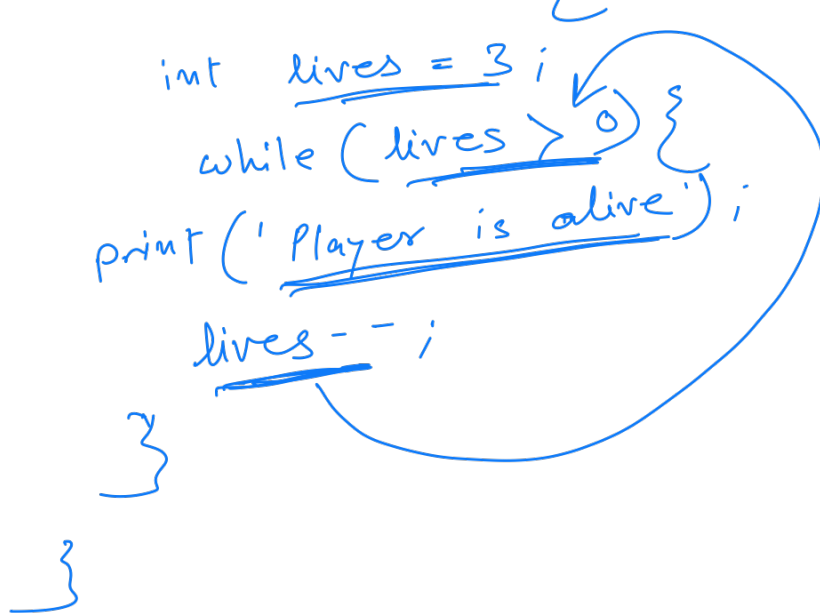
1
2
3
4
5

b) infinite Loop

```
void main() {  
    while (true) {  
        print('Running....');  
    }  
}
```

c) Real life Example

```
void main() {  
    int lives = 3 i  
    while (lives > 0) {  
        print('Player is alive');  
        lives--;  
    }  
}
```



d) Sum of digits

% /

void main() {

int number = 12345;

int sum = 0;

while (number > 0) {

sum += number % 10;

number /= 10;

}

print('Sum of digits : %sum');

}

sum = sum + num % 10

5 = 0 + 12345 % 10

5 = 0 + 5

number = number / 10

1234 = 12345 / 10

sum = sum + num % 10

9 = 5 + 4

num = num / 10

1234 = 12345 / 10

sum = sum + num % 10

12 = 9 + 3

num = 12

sum = sum + num % 10

14 = 12 + 2

num = num / 10

sum = 15

3. Do-while Loop

Syntax:

do {

code

} while (condition);

void main() {

int i = 10;

do {

print(i);

} while (i <= 5);

}

* Step up (1-5)

void main() {

int i = 1;

do {

print(i);

i++

} while (i <= 5);

}

1
2
3
4
5

```

void void main() {
    int i = 10;
    do {
        print('This will run once: $i');
    } while (i <= 5);
    // while (i <= 5) {
        print('This won't run');
    }
}

```

4. for-in loop

```

void main() {
    List<String> fruits = ['Apple', 'Mango', 'Banana'];
    for (var fruit in fruits) {
        print(fruit);
    }
}

```

// Apple
// Mango
// Banana

```

void void main() {
    List<int> numbers = [10, 20, 30, 20, 10];
    for (var number in numbers) {
        print(number);
    }
}

```

10
20
30
20
10

```

void main() {
    Set<String> fruits = {
        'Apple', 'Mango', 'Banana' };
    for (var fruit in fruits) {
        print(fruit);
    }
}

```

Apple
Mango
Banana

```

void main() {
    Set<int> numbers = { 10, 20, 30, 20, 10 };
    for (var number in numbers) {
        print(number);
    }
}

```

10
20
30

or Map

```

void main() {
    Map<String, int> marks = {
        'John' : 80,
        'Alice' : 90,
        'Bob' : 75
    };
    for (var key in marks.keys) {
        print(key);
    }
    for (var value in marks.values) {
        print(value);
    }
    for (var entry in marks.entries) {
        print(" {entry.key} : {entry.value} ");
    }
}

```

John
Alice
Bob

80
90
75

John : 80
Alice : 70
Bob : 75

* Real Life Example

```
void main() {  
    Map<String, int> students = {  
        'Rahim' : 85,  
        'Karrim' : 45,  
        'Sakib' : 70  
    };  
    for (var student in students.entries) {  
        if (student.value >= 50) {  
            print(' $ {student.Key} : Passed ');  
        } else {  
            print(' $ {student.Key} : Failed ');  
        }  
    }  
}
```

// Rahim : Passed
// Karrim : Failed
// Sakib : Passed //

5. forEach()

```
void main() {  
    List<String> fruits = ["Apple", "Mango", "Banana"];  
    {fruits.forEach((fruit) {  
        print(fruit);  
    });  
    // fruits.forEach((fruit) => print(fruit));  
}
```

6. Nested loop

```

void main() {
    for (int i = 1; i <= 3; i++) {
        for (int j = 1; j <= 3; j++) {
            print("i = %i, j = %j");
        }
    }
}

```

i = 1, j = 1
 i = 1, j = 2
 i = 1, j = 3
 i = 2, j = 1
 i = 2, j = 2
 i = 2, j = 3
 i = 3, j = 1
 i = 3, j = 2
 i = 3, j = 3

* Multiplication Table (1-5)

```

void main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 10; j++) {
            print("%i x %j = %i * %j");
        }
        print("-----");
    }
}

```

1 x 1 = 1
 1 x 2 = 2

 1 x 10 = 10

 2 x 1 = 2
 2 x 2 = 4

 2 x 10 = 20

7. Loop Control Statement

- Break
- Continue

• break


```

void main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 5) {
            break;
        }
        print(i);
    }
}

```

1
2
3
4

• Continue

```

void main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 3 || i == 4 || i == 7 || i == 9) {
            continue;
        }
        print(i);
    }
}

```

1
2
5
6
8
10

8. Examples

1) Sum of Numbers 1 to 10

```

void main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++) {
        sum = sum + i;
    }
    print("Sum = $sum");
}

```

$1 = 0 + 1$
 $3 = 1 + 2$
 $6 = 3 + 3$
 $10 = 6 + 4$
 $15 = 10 + 5$
 $21 = 15 + 6$
 $28 = 21 + 7$
 $36 = 28 + 8$
 $45 = 36 + 9$
55 = 45 + 10

* Factorial

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

```

void main() {
    int number = 5;
    int factorial = 1;
    for (int i = 1; i <= number; i++) {
        factorial *= i;
    }
    print(factorial);
}

```

$1 = 1 \times 1$
 $2 = 1 \times 2$
 $6 = 2 \times 3$
 $24 = 6 \times 4$
 $120 = 24 \times 5$

* find the largest number in list

[10, 25, 5, 40, 15];

```

void main() {
    List<int> numbers = [10, 25, 5, 40, 15];
    int largest = numbers[0];
    for (int number in numbers) {
        if (number > largest) {
            largest = number;
        }
    }
    print(largest);
}

```

9. Patterns

	① Column Numbers			
1	2	3	4	
1	2	3	4	
1	2	3	4	
1	2	3	4	

1	2	3
1	2	3
1	2	3

1	2	3	4	r
1	2	3	4	r
1	2	3	4	r
1	2	3	4	r
1	2	3	4	r

```
import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
```

```
  if (no == null) {
    print('Invalid Input!');
    return;
  }
```

```
  for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= no; j++) {
      stdout.write('$j ');
    }
```

```
    print(' ');
  }
```

1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

2. Pattern (Row Number Pattern)

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

1 1 1

2 2 2

3 3 3

import 'dart:io';

void main() {

Stdout.write('Enter a number of rows: ');

int? no = int.tryParse(Stdin.readLineSync()!);

if (no == null) {

print("Invalid Input!");

return;

}

for (int i = 1; i <= no; i++) {

for (int j = 1; j <= no; j++) {

Stdout.write('\$i ');

}

print(' ');

}

}

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4

3. Pattern (Reverse Column Number)

4 3 2 1

4 3 2 1

4 3 2 1

4 3 2 1

3 2 1

3 2 1

3 2 1

```

import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
  if (no == null) {
    print('Invalid Input!');
    return;
  }

  for (int i = 1; i <= no; i++) {
    for (int j = no; j >= 1; j--) {
      stdout.write('$j ');
    }
    print('');
  }
}

```

4 3 2 1
4 3 2 1
4 3 2 1
4 3 2 1

4. Reverse Row Number Pattern

```

4 4 4 4
3 3 3 3
2 2 2 2
1 1 1 1

```

```

import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
  if (no == null) {
    print('Invalid Input!');
    return;
  }
}

```

```

for (int i = no; i >= 1; i--) {
  for (int j = 1; j <= no; j++) {
    stdout.write('$i ');
  }
  print('');
}

```

4 4 4 4
3 3 3 3
2 2 2 2
1 1 1 1

5. Floyd's square Pattern

```

1  2  3  4
5  6  7  8
9 10 11 12
13 14 15 16

```

```

1  2  3
4  5  6
7  8  9

```

```

import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
  if (no == null) {
    print('Invalid Input!');
    return;
  }

  int ch = 1;
  for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= no; j++) {
      String fch = ch < 10 ? '0$ch' : '$ch';
      stdout.write(fch);
      ch++;
    }
    print("");
  }
}

```

```

1  2  3
4  5  6
7  8  9

```

6. Reverse Floyd's square Pattern

```

16 15 14 13
12 11 10 9
8 7 6 5
4 3 2 1

```

```

9 8 7
6 5 4
3 2 1

```

```

import 'dart:io';

void main() {
  Stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync());
  if (no == null) {
    print('Invalid Input!');
    return;
  }

  int ch = no * no;
  for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= no; j++) {
      String fch = ch < 10 ? '0$ch' : '$ch';
      Stdout.write('$fch');
      ch--;
    }
    print("");
  }
}

```

```

09 08 07
06 05 04
03 02 01

```

7. Hollow square Pattern by any character

```

* * * *
*       *
*       *
*       *
* * * *

```

```

x x x x
x       x
x       x
x       x
x x x x

```

```

import 'dart:io';

void main() {
  Stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync());
  Stdout.write('Enter a character: ');
  String input = stdin.readLineSync() ?? '';
  String ch = input.isEmpty ? '*' : input[0];

  if (no == null) {
    print('Invalid Input!');
  }
}

```

```

    }
    for (int i = 1; i <= no; i++) {
        for (int j = 1; j <= no; j++) {
            if (i == 1 || i == no || j == 1 || j == no) {
                stdout.write('$ch');
            } else {
                stdout.write(' ');
            }
        }
        print('\n');
    }
}

```

③ #

#

- -

#

8. Binary Pattern

```

0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0

```

```

0 1 0
1 0 1
0 1 0

```

```

import 'dart:io';

void main() {
    stdout.write('Enter a number of row: ');
    int? no = int.tryParse(stdin.readLineSync());
    if (no == null) {
        print('Invalid Input!');
        return;
    }
}

```

```

for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= no; j++) {
        stdout.write('${(i+j)%2} ');
    }
    print('\n');
}

```



```

}
}

```

③

0	1	0
1	0	1
0	1	0

9. Right Angle Triangle (Number)

```

1
1 2
1 2 3
1 2 3 4

```

```

import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
  if (no == null) {
    print('Invalid Input!');
    return;
  }
  for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= i; j++) {
      stdout.write('$j ');
    }
    print(' ');
  }
}

```

③

```

1
1 2
1 2 3

```

10.

```

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```

```

import 'dart:io';

void main() {
  stdout.write('Enter a number of rows: ');
  int? no = int.tryParse(stdin.readLineSync()!);
  if (no == null) {
    print('Invalid Input!');
    return;
  }

```

```

  for (int i = 1; i <= no; i++) {
    for (int j = 1; j <= i; j++) {
      stdout.write('$i ');
    }

```

```

  }
  print('');
}

```

```

1
2 2
3 3 3

```

H/w

- 1) Print numbers from 1 to 20 using a for loop
- 2) Print all even numbers from 1 to 50
- 3) Find the sum of numbers from 1 to 100
- 4) Print a multiplication table of 7
- 5) Find the factorial of 6.